



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to MARRIYAMMAL AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

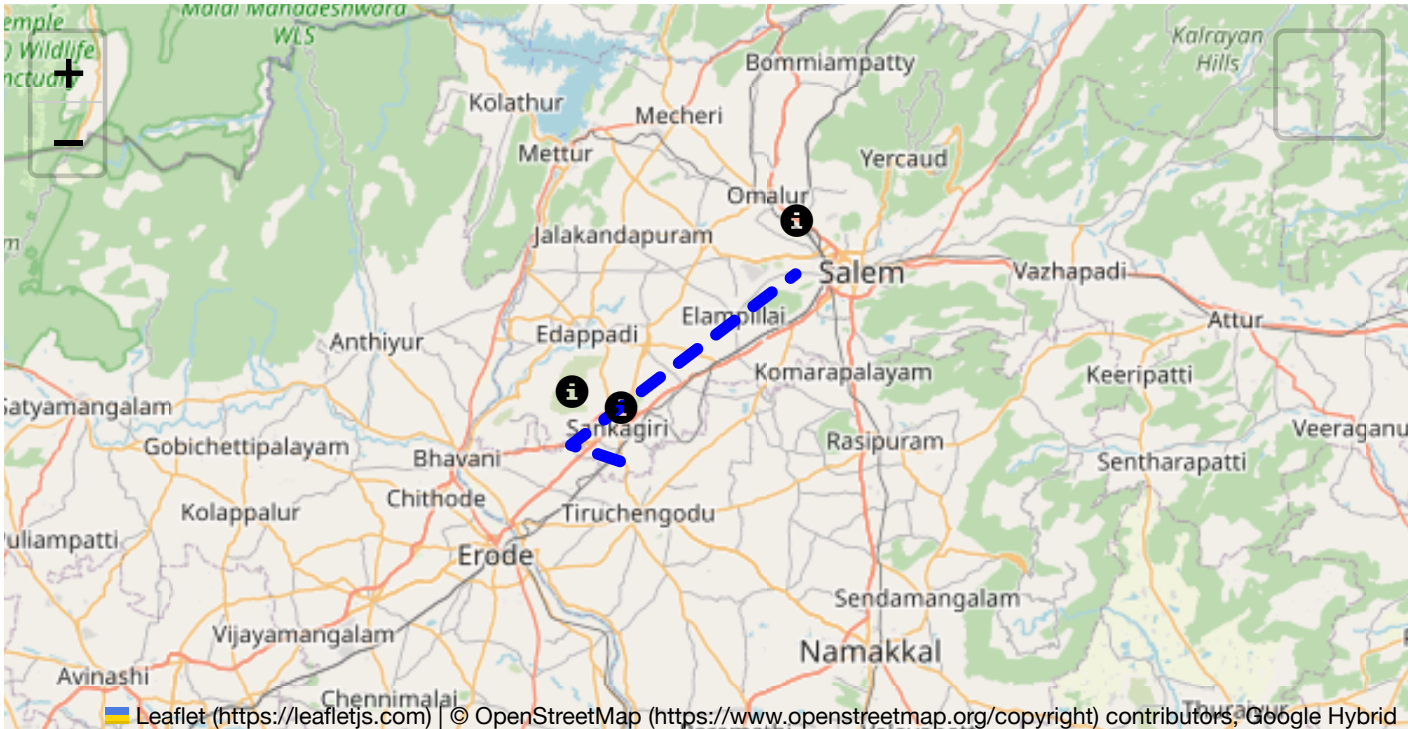
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

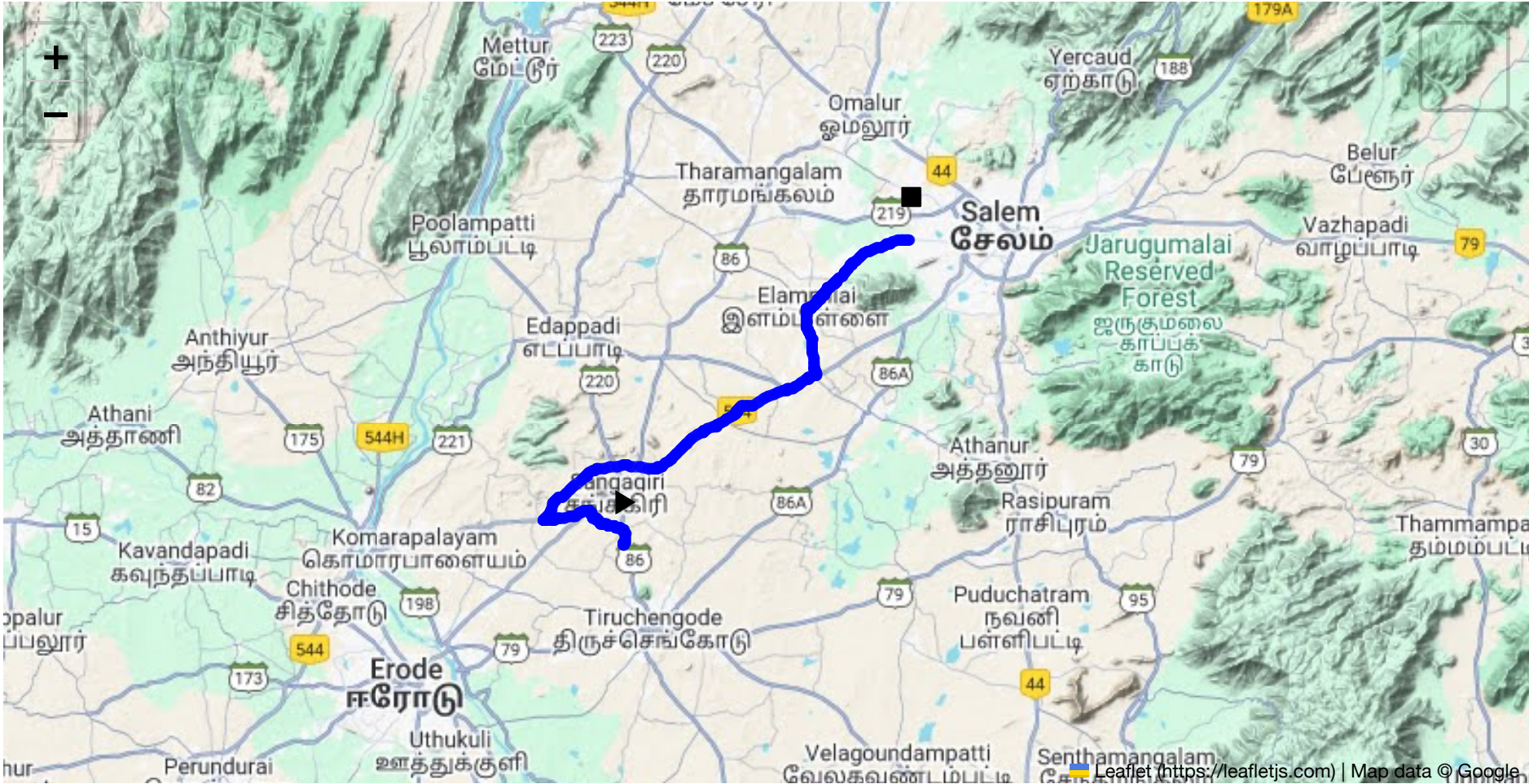
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 49.57 km
Estimated Duration: 1.1 hours
Adjusted Duration (Heavy Vehicle): 1.4 hours

Start: (11.4381, 77.8734)
End: (11.653496, 78.079255)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route spans approximately 49.57 kilometers, starting from Sangagiri and passing through Padaiveedu via Bhavani Main Road before reaching the destination in Salem, Tamil Nadu. This route covers a combination of rural and semi-urban landscapes, providing a mix of two-lane state highways and local roads.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- Temperature:** Typically hot, especially in summer (March to June), with temperatures reaching above 40°C.
- Monsoon Season:** Heavy rains occur from June to September, which might lead to occasional flooding and slippery roads.
- Fog:** During winter months (December to February), there might be occasional fog early in the morning.

3. Analysis of Traffic Patterns

- Peak Hours:** Morning (8:00-10:00 AM) and evening (5:00-7:00 PM) are peak times due to local commuters.
- Congestion-prone Areas:** The village areas near Padaiveedu and entry points towards Salem, particularly near the bus stand, are prone to traffic snarls.

4. Assessment of Road Quality and Infrastructure

- **State Highway:** Primarily well-maintained, but some segments might have potholes, especially after the monsoon.
- **Rural Roads:** Could be narrow with sharp turns, requiring cautious navigation for heavy vehicles.
- **Signage:** Adequate in most parts, but some key local turns might lack sufficient signs.

5. Suggestions for Alternative Routes for Emergencies

- **Alternative Route 1:** Via Erode-Salem road, it's longer but well-maintained and with better emergency services coverage.
- **Alternative Route 2:** Through NH44, offers quicker access to major facilities but might be subject to tolls.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Night Travel Restrictions:** Some stretches restrict heavy vehicles during night hours.
- **Material Specific Regulations:** Comprehensive checks at check-posts and requirement for specific permits depending on the type of material.

7. Overview of Historical Incidents

- **Accidents:** There have been incidents involving heavy vehicles due to speeding and poor road visibility, especially near village intersections.
- **Hazmat Spills:** Rare, but incidents have generally involved improper handling or regulatory non-compliance.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Areas:** Proximity to agricultural lands requires cautious handling of spills.
- **Pollution Regulations:** Strict adherence to emission control standards is necessary, especially near populated settlements.

9. Analysis of Communication Coverage

- **Coverage:** Mostly good with 4G services available, but sporadic dead zones could exist in forested or rural patches.
- **Dead Zones:** Primarily in more rural segments, especially transitioning between village areas.

10. Estimated Emergency Response Times

- **Sangagiri to Padaiveedu:** Approximately 20-30 minutes.
- **Padaiveedu to Salem:** Could extend to an hour due to distance and traffic.

11. Overall Summary of Risk Assessment

This route involves moderate risk for heavy vehicles carrying hazardous materials due to variable road conditions and weather hazards. Proper planning around peak hours, monitoring weather forecasts, and rigorous vehicle checks are recommended to minimize risks. Environmental and safety regulations must

be followed diligently. Communication and coordination with local emergency services are key, and alternative route knowledge is crucial in case of roadblocks or other impediments.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.45353, 77.85697	30 KM/Hr
5	Turn	Medium	11.45839, 77.85143	30 KM/Hr
6	Turn	Medium	11.46307, 77.84941	30 KM/Hr
7	Turn	Medium	11.46316, 77.84921	30 KM/Hr
8	Turn	High	11.45542, 77.81725	15 KM/Hr
9	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
10	Turn	High	11.55839, 78.01288	15 KM/Hr
11	Turn	Medium	11.60320, 78.00504	30 KM/Hr
12	Turn	Medium	11.60480, 78.00617	30 KM/Hr
13	Turn	Medium	11.60481, 78.00747	30 KM/Hr
14	Turn	High	11.60552, 78.00827	15 KM/Hr
15	Turn	Medium	11.60546, 78.00846	30 KM/Hr
16	Turn	Medium	11.60751, 78.01086	30 KM/Hr
17	Turn	Medium	11.61152, 78.01356	30 KM/Hr
18	Turn	High	11.61170, 78.01516	15 KM/Hr
19	Turn	Medium	11.61418, 78.01669	30 KM/Hr
20	Turn	Medium	11.61440, 78.01791	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vembadithalam Government General Hospital	11.5683, 78.01211	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
4	hospital	Subam Hospital	11.6069532, 78.0042593	30 km/h	Medium
5	hospital	SR Hospital	11.6075985, 78.0070051	30 km/h	Medium
6	hospital	Nesam Medical Centre and Hospital	11.6078647, 78.0073456	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	school	Jyothi Vidyalaya Matric HR Sec School	11.6194482, 78.0212238	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45839, 77.85143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46316, 77.84921



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60320, 78.00504



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60480, 78.00617



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60481, 78.00747



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60552, 78.00827



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60546, 78.00846



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60751, 78.01086



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61152, 78.01356



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.61170, 78.01516



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61418, 78.01669



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61440, 78.01791
